

Tom Silver

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Education

Ph.D. Massachusetts Institute of Technology

Computer Science (EECS)

Advised by Leslie Kaelbling and Joshua Tenenbaum

Thesis: *Neuro-Symbolic Learning for Bilevel Robot Planning*

Cambridge, MA
May 2020 – May 2024

S.M. Massachusetts Institute of Technology

Computer Science (EECS)

Thesis: *Few-Shot Bayesian Imitation Learning with Logical Program Policies*

GPA: 5.0/5.0

Cambridge, MA
Aug 2018 – May 2020

A.B. Harvard College

Computer Science and Mathematics

Thesis: *Luna: A Game-Based Rating System for Artificial Intelligence*

Magna cum laude with highest honors

Cambridge, MA
Aug 2012 – May 2016

Employment

Princeton University

Assistant Professor, ECE Department

Principal Investigator, Princeton Robot Planning and Learning (PRPL)

Associated Faculty, Center for Statistics and Machine Learning

Core Faculty, Robotics

Princeton, NJ
July 2025 – Present

Cornell University

Postdoctoral Fellow

EmPRISE Lab (PI: Tapomayukh "Tapo" Bhattacharjee)

CS Department

Ithaca, NY
July 2024 – June 2025

Boston Dynamics AI Institute

Research Intern

Manager: Jenny Barry

Cambridge, MA
May 2023 – May 2024

Google Brain Robotics

Research Intern

Managers: Shane Gu and Saminda Abeyruwan

Mountain View, CA (Remote)
Jun – Aug 2021

Vicarious AI

Research Engineer; Researcher

Union City, CA
Jul 2016 – Aug 2018

Google

Software Engineering Intern

Cambridge, MA
Jun – Aug 2014

Conference Publications

* Equal contribution † Students for whom I was the primary supervisor

1. Li, B., **Silver, T.**, Scherer, S., Gray, A. Bilevel Learning for Bilevel Planning. *Robotics: Science and Systems (RSS) 2025*.
2. Jenamani, R., **Silver, T.**, Dodson, B., Tong, S., Song, A., Yang, Y., Liu, Z., Howe, B., Whitneck, A., Bhattacharjee, T. FEAST: A Flexible Mealtime-Assistance System Towards In-the-Wild Personalization. *Robotics: Science and Systems (RSS) 2025*. **Best paper award (Top 1)**.
3. Liang, Y., Kumar, N., Tang, H., Weller, A., Tenenbaum, J., **Silver, T.**, Henriques, J., Ellis, K. VisualPredicator: Learning Abstract World Models with Neuro-Symbolic Predicates for Robot Planning. *International Conference on Learning Representations (ICLR) 2025*. **Spotlight presentation (Top 5%)**.
4. Ye, R., Chen, S., Yan, Y., Yang, J., Ge, C., Barreiros, J., Tsui, K., **Silver, T.**, Bhattacharjee, T. CART-MPC: Coordinating Assistive Devices for Robot-Assisted Transferring with Multi-Agent MPC. *ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2025*.
5. Liu, Z., Ju, Y., Da, Y., **Silver, T.**, Thakkar, P., Li, J., Guo, J., Dimitropoulou, K., Bhattacharjee, T. GRACE: Generalizing Robot-Assisted Caregiving with User Functionality Embeddings. *ACM/IEEE International Conference on Human-Robot Interaction (HRI) 2025*.
6. Kumar, N.*†, **Silver, T.***, McClinton, W.†, Zhao, L., Proulx, S., Lozano-Perez, T., Kaelbling, L., Barry, J. Practice Makes Perfect: Planning to Learn Skill Parameter Policies. *Robotics: Science and Systems (RSS) 2024*.
7. **Silver, T.**, Dan, S., Srinivas, K., Tenenbaum, J., Kaelbling, L., Katz, M. Generalized Planning in PDDL Domains with Pretrained Large Language Models. *AAAI Conference on Artificial Intelligence (AAAI) 2024*.
8. Kumar, N.*†, McClinton, W.*†, Chitnis, R., **Silver, T.**, Lozano-Perez, T., Kaelbling, L. Learning Efficient Abstract Planning Models that Choose What to Predict. *Conference on Robot Learning (CoRL) 2023*.
9. Li, A.†, **Silver, T.** Embodied Active Learning of Relational State Abstractions for Bilevel Planning. *Conference on Lifelong Learning Agents (CoLLAs) 2023*. **Oral presentation (Top 12)**.
10. **Silver, T.***, Chitnis, R.*, Kumar, N.†, McClinton, W.†, Lozano-Perez, T., Kaelbling, L., Tenenbaum, J. Predicate Invention for Bilevel Planning. *AAAI Conference on Artificial Intelligence (AAAI) 2023*. **Oral presentation**.
11. **Silver, T.**, Athalye, A.†, Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. Learning Neuro-Symbolic Skills for Bilevel Planning. *Conference on Robot Learning (CoRL) 2022*.
12. Chitnis, R.*, **Silver, T.***, Tenenbaum, J., Kaelbling, L., Lozano-Perez, T. Learning Neuro-Symbolic Relational Transition Models for Bilevel Planning. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022*.
13. Yang, R.*†, **Silver, T.***, Curtis, A., Lozano-Perez, T., Kaelbling, L. PG3: Policy-Guided Planning for Generalized Policy Generation. *International Joint Conference on Artificial Intelligence (IJCAI) 2022*.

14. Gehring, C.*, Asai, M.*, Chitnis, R., **Silver, T.**, Kaelbling, L., Sohrabi, S., Katz, M. Reinforcement Learning for Classical Planning: Viewing Heuristics as Dense Reward Generators. *International Conference on Automated Planning and Scheduling (ICAPS)* 2022.
15. Curtis, A., **Silver, T.**, Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. Discovering State and Action Abstractions for Generalized Task and Motion Planning. *AAAI Conference on Artificial Intelligence (AAAI)* 2022.
16. **Silver, T.***, Chitnis, R.*, Tenenbaum, J., Kaelbling, L., Lozano-Perez, T. Learning Symbolic Operators for Task and Motion Planning. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2021. **Best paper finalist (Top 5).**
17. **Silver, T.***, Chitnis, R.*, Curtis, A., Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. Planning with Learned Object Importance in Large Problem Instances Using Graph Neural Networks. *AAAI Conference on Artificial Intelligence (AAAI)* 2021.
18. Chitnis, R.*, **Silver, T.***, Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. GLIB: Efficient Exploration for Relational Model-based Reinforcement Learning via Goal-Literal Babbling. *AAAI Conference on Artificial Intelligence (AAAI)* 2021.
19. Chitnis, R.*, **Silver, T.***, Kim, B., Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. CAMPs: Learning Context-Specific Abstractions for Efficient Planning in Factored MDPs. *Conference on Robot Learning (CoRL)* 2020. **Plenary talk (Top 20).**
20. Zhi-Xuan, T., Mann J. L., **Silver, T.**, Tenenbaum, J., Mansinghka, V. K. Online Bayesian Goal Inference for Boundedly-Rational Planning Agents. *Conference on Neural Information Processing Systems (NeurIPS)* 2020.
21. **Silver, T.**, Allen, K., Lew, A., Kaelbling, L., Tenenbaum, J. Few-Shot Bayesian Imitation Learning with Logical Program Policies. *AAAI Conference on Artificial Intelligence (AAAI)* 2020. Earlier versions at RLDM 2019 and ICLR SPiRL Workshop 2019.
22. Loula, J., Allen, K., **Silver, T.**, Tenenbaum, J. Learning Constraint-Based Planning Models from Demonstrations. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2020.
23. Xia, V.*, Wang, Z.*, Allen, K., **Silver, T.**, Kaelbling, L. Learning Sparse Relational Transition Models. *International Conference on Learning Representations (ICLR)* 2019.
24. Loula, J., **Silver, T.**, Allen, K., Tenenbaum, J. Discovering a Symbolic Planning Language from Continuous Experience. *Annual Meeting of the Cognitive Science Society (CogSci)* 2019.
25. Stark, M., Schlegel, A., Wendelken, C., Park, D., Purdy, E., **Silver, T.**, Phoenix, S., George, D. Behavior is Everything – Towards Representing Concepts with Sensorimotor Contingencies. *AAAI Conference on Artificial Intelligence (AAAI)* 2018.
26. Kansky, K., **Silver, T.**, Mely, D. A., Eldawy, M., Lazaro-Gredilla, M., Lou, X., Dorfman N., Sidor S., Phoenix S., George, D. Schema Networks: Zero-Shot Transfer with a Generative Causal Model of Intuitive Physics. *International Conference on Machine Learning (ICML)* 2017.

Journal Publications

27. Garrett, C.R., Chitnis, R., Holladay, R., Kim, B., **Silver, T.**, Kaelbling, L., Lozano-Perez, T. Integrated Task and Motion Planning. *Annual Review of Control, Robotics, and Autonomous Systems*. Vol. 4 2021.
28. Colubri, A.*, **Silver, T.***, Fradet, T., Retzepi, K., Fry, B., Sabeti, P. Transforming Clinical Data into Actionable Prognosis Models: Machine-Learning Framework and Field-Deployable App to Predict Outcome of Ebola Patients. *PLoS Neglected Tropical Diseases* 2016.

Workshop Publications

29. Liu, Y, Li, B., Eysenbach, B., **Silver, T.** SLAP: Shortcut Learning for Abstract Planning. *R3: Reasoning for Robust Robot Manipulation in the Open World @ RSS* 2025.
30. Zhao, L., McClinton, W., Curtis, A., Kumar, N, **Silver, T.**, Kaelbling, L, Wong, L. Seeing is Believing: Planning to Perceive with Foundation Models and Act Under Uncertainty. *Workshop on Robot Planning in the Era of Foundation Models @ RSS* 2025.
31. **Silver, T.***, Hariprasad, V.*[†], Shuttleworth, R.*[†], Kumar, N.[†], Lozano-Perez, T., Kaelbling, L. PDDL Planning with Pretrained Large Language Models. *Workshop on Foundation Models for Decision Making (FMDM) @ NeurIPS* 2022.
32. **Silver, T.***, Chitnis, R.*, Kumar, N.[†], McClinton, W.[†], Lozano-Perez, T., Kaelbling, L., Tenenbaum, J. Inventing Relational State and Action Abstractions for Effective and Efficient Bilevel Planning. *Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM)* 2022. Short version of “Predicate Invention for Bilevel Planning” (AAAI 2023). **Spotlight talk.**
33. Zeng, C.[†], **Silver, T.** Learning Search Guidance from Failures with Eliminate Edge Sets. *Workshop on Bridging the Gap Between AI Planning and Reinforcement Learning (PRL) @ ICAPS* 2021.
34. **Silver, T.**, Chitnis, R. PDDLgym: Gym Environments from PDDL Problems. *Workshop on Bridging the Gap Between AI Planning and Reinforcement Learning (PRL) @ ICAPS* 2020.
35. **Silver, T.***, Chitnis, R.*, Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. Learning Skill Hierarchies from Predicate Descriptions and Self-Supervision. *Workshop on Generalization in Planning (GenPlan) @ AAAI* 2020.
36. Loula, J., **Silver, T.**, Allen, K., Tenenbaum, J. Learning Models for Mode-based Planning. *Workshop on Generative Modeling and Model-Based Reasoning for Robotics and AI (MBRL) @ ICML* 2019.

Preprints and Others

37. **Silver, T.**, Jenamani, R., Liu, Z., Dodson, B., Bhattacharjee, T. Coloring Between the Lines. *arXiv* 2025.
38. **Silver, T.***, Allen, K.*, Tenenbaum, J., Kaelbling, L. Residual Policy Learning. *arXiv* 2018.
39. **Silver, T.** Luna: A Game-Based Rating System for Artificial Intelligence. *Undergraduate thesis with Professor Stuart M. Shieber*, Harvard College 2016.

Teaching

ECE 531 / COS 531: Robot Planning Meets Machine Learning Instructor; Course Creator	Princeton University Aug – Dec 2025
6.s058 / 16.420: Representation, Inference and Reasoning in AI Co-instructor	MIT Aug – Dec 2021
6.882: Structured Models for AI Teaching Assistant	MIT Aug – Dec 2020
CS 121: Theory of Computation Head Teaching Fellow; Teaching Fellow	Harvard College Sep – Dec 2014; Sep – Dec 2015
CS 20: Discrete Math Head Teaching Fellow; Teaching Fellow	Harvard College Jan – May 2015; Jan – May 2016

Invited Talks

RSS 2025 Workshop: Robot Planning in the Era of Foundation Models (FM4RoboPlan)	Jun 2025
Summer School on Neurosymbolic Programming	Jun 2024
Cornell Tech (Rush Group)	Nov 2023
Cornell University (Bhattacharjee Group)	Oct 2023
Yale University (Scassellati Group)	Sep 2023
Princeton University (Griffiths Group)	Mar 2023
Rutgers University (CS Department Colloquium)	Mar 2023
Columbia University (Song Group)	Feb 2023
George Mason University (Controls and Robotics Seminar)	Feb 2023
AAAI AI & Robotics Bridge Session	Feb 2023
Oxford University (CS Department Seminar)	Jan 2023
DeepMind London (Shanahan Group)	Jan 2023
New York University (Pinto Group)	Jan 2023
Massachusetts Institute of Technology (Andreas Group)	Nov 2022
Northeast Robotics Colloquium (NERC) (Poster)	Oct 2022
University of New Hampshire (Ruml Group)	Sep 2022
Meta Reality Labs Research (Desai Group)	Sep 2022
Stanford University (BEHAVIOR Group)	Sep 2022
Technical University of Berlin (TU Berlin) (Toussaint Group)	Dec 2021
CogSci 2021 Workshop: Minds At Play	Jul 2021
Allen Institute for AI (AI2)	Mar 2021
Delft University of Technology (TU Delft) (Oliehoek Group)	Feb 2021
Brown University (Konidaris, Tellex, Littman Groups)	Jan 2021
Arizona State University (Srivastava Group)	Jan 2021
Brown University (Konidaris, Tellex, Littman Groups)	Mar 2020

Fellowships, Honors, and Awards

Outstanding MIT UROP Mentor Awards (Nominated)	Apr 2023
IJCAI-ECAI 2022 Distinguished PC (Top 3% Reviewer)	Sep 2022
MIT EECS Hazen Teaching Award	June 2022
NSF Graduate Research Fellowship	Aug 2018 – May 2023
MIT Stata Family Presidential Fellowship	Aug 2018 – May 2019
Blumberg Creative Science Prize (Mather House, Harvard)	May 2016
Derek Bok Certificate of Distinction in Teaching (Harvard)	Jan 2015, May 2015, Jan 2016
Harvard College PRISE Fellowship	May – June 2013

Research Science Institute (RSI)

May – June 2011

Workshop and Competition Organizing Committees

PhyRC: Physical Robotic Caregiving Challenge @ ICRA 2025

Co-organizers: Ruolin Ye, Shuaixing Chen, Justin Guo, Martin Leroux, Binit Shah, and Tapomayukh Bhattacharjee

Atlanta, GA

May 2025

Learning Effective Abstractions for Planning (LEAP) @ CoRL 2023

Co-organizers: Naman Shah, Eric Rosen, David Paulius, Beomjoon Kim, Georgia Chalvatzaki

Atlanta, GA

Nov 2023

Learning for Task and Motion Planning @ RSS 2023

Co-organizers: Danfei Xu, Danny Driess, Jeannette Bohg, Rohan Chitnis, Shuo Cheng, Zhutian Yang

Daegu, Korea

Jul 2023

Learning, Perception, and Abstraction for Long-Horizon Planning @ CoRL 2022

Co-organizers: Gregory Stein, Jana Kosecka, Rohan Chitnis, Yezhou Yang

Auckland, NZ

Dec 2022

Advisees

Isabel Liu (Princeton Undergrad)

Fall 2024 - Present

Nishanth Kumar (MIT PhD Student)

Fall 2021 – May 2024

Willie McClinton (MIT PhD Student)

Fall 2021 – May 2024

Ryan Yang (MIT Undergrad → Applied Intuition)

Spring 2021 – May 2024

Reece Shuttleworth (MIT Undergrad)

Summer 2022 – Spring 2023

Ashay Athalye (MIT Master's Student → Boston Dynamics AI)

Fall 2021 – Spring 2023

Lilian Luong (MIT Master's Student → Applied Intuition)

Fall 2022 – Spring 2023

Jagdeep Bhatia (MIT Undergrad → Berkeley PhD Student)

Fall 2021 – Spring 2023

Amber Li (MIT Master's Student → CMU PhD Student)

Fall 2021 – Fall 2022

Varun Hariprasad (RSI → MIT Undergrad)

Summer 2022

Abraham Mitchell (University of Arkansas Undergrad, MSRP)

Summer – Fall 2022

Wester J Aldarondo-Torres (UPR Undergrad, MSRP)

Summer – Fall 2022

Catherine Zeng (Harvard Undergrad → CMU PhD Student)

Fall 2020 – Spring 2022

Shariqah N Hossain (MIT Undergrad → InterSystems)

Fall 2020 – Spring 2021

Reviewing

AAAI Conference on Artificial Intelligence (AAAI)

2021, 2022, 2023, 2024

Autonomous Robots

2024

Conference on Robot Learning (CoRL)

2022, 2023, 2024, 2025

Frontiers in Robotics and AI

2024

Generalization in Planning (GenPlan Workshop)

2021, 2022, 2023

International Conference on Human-Robot Interaction (HRI)

2025

International Conference on Learning Representations (ICLR)

2021, 2022

International Conference on Machine Learning (ICML)

2022, 2023

International Conference on Robotics and Automation (ICRA)

2021, 2022, 2023, 2025

International Joint Conference on Artificial Intelligence (IJCAI)

2022, 2023

International Journal of Robotics Research (IJRR)

2025

International Conference on Intelligent Robots and Systems (IROS)

2023, 2024

International Symposium on Robotics Research (ISRR)	2022
Journal of Artificial Intelligence Research (JAIR)	2022
Conference on Neural Information Processing Systems (NeurIPS)	2021, 2022, 2023, 2025
Planning and Robotics (PlanRob @ ICAPS)	2023
Programmatic Representations for Agent Learning (PRAL @ ICML)	2025
Bridging Planning and Reinforcement Learning (PRL @ IJCAI)	2023
Robotics and Automation Letters (RA-L)	2021, 2022, 2023, 2025
Reinforcement Learning Conference (RLC)	2025
Robotics: Science and Systems (RSS)	2024, 2025
Transactions on Human-Robot Interaction (THRI)	2025
Transactions on Machine Learning Research (TMLR)	2022, 2023
